Supplementary Material for LSMRA: LLM-Enhanced Semantic Multi-Agent Risk Analysis Framework for Interbank Credit Rating

Yuanzhong Li, Yuxi Yan, Kun Zhu, Nana Zhang, Zhiquan Liu, and Changjun Jiang

***Abstract*—This supplementary material provides the detailed algorithms and the semantic embedding prompt used by LSMRA. Notation follows the main paper.**

I. Algorithms

A. End-to-End Pipeline

Algorithm 1 End-to-End LSMRA Pipeline

Input: target quarter t; detection window $T_{\text{win}} = 3$; attack rate $\rho \in [0, 0.25]$; semantic/risk enable flags $s_{\text{sem}}, s_{\text{risk}} \in \{0, 1\}$

Output: rating predictions $\hat{Y}$; refined anomaly set S_2; semantic inconsistency scores Ω; risk scores ψ

Load features $\{X^{(\tau)}\}$, adjacencies $\{A^{(\tau)}\}$, and ratings $\{y_i^{(\tau)}\}$ for $\tau \in \{t-2, t-1, t\}$; if $\rho > 0$, load the attacked snapshots $(\tilde{X}^{(\tau)}, \tilde{A}^{(\tau)})$;
Phase 1: Multi-Agent Suspicious Screening
Obtain S_1, $\{s(i)\}$, and $\{c_k(i)\}$ by Algorithm 2$(\{X^{(\tau)}\}, \{A^{(\tau)}\}, \{F_g\}, \{y_i^{(\tau)}\})$;
Phase 2: LLM-Enhanced Semantic Verification
if $s_{\text{sem}} = 1$ **then**
    Obtain S_2 and Ω by Algorithm 3$(S_1, X^{(t)}, A^{(t)}, \{y_i^{(t)}\})$;
else
    $S_2 \leftarrow S_1$; $\Omega \leftarrow \mathbf{0}$;
end if
Phase 3: Attention-Guided Risk Propagation
if $s_{\text{risk}} = 1$ **then**
    Obtain ψ by Algorithm 4$(\Omega, S_2, A^{(t)}, \{y_i^{(\tau)}\})$;
else
    $\psi \leftarrow \mathbf{0}$;
end if
Phase 4: Security-Aware Rating Prediction
Build anomaly-aware features X' from group summaries, multi-agent screening evidence, Ω, and ψ;
$\hat{Y} \leftarrow \texttt{AnomalyAwareGAT}(X', A^{(t)})$;
return $\hat{Y}$, S_2, Ω, ψ.

B. Multi-Agent Suspicious Screening

Algorithm 2 Multi-Agent Suspicious Screening

Input: multi-quarter financial features $\{X^{(\tau)}\}_{\tau \in \{t-2, t-1, t\}}$ and adjacencies $\{A^{(\tau)}\}$; feature-group index sets $\{F_g\}_{g=1}^{G_{\text{grp}}}$; ratings $\{y_i^{(\tau)}\}$ (optional)

Output: candidate set S_1; fused score vector $\{s(i)\}_{i=1}^{N}$; agent-wise confidence vectors $\{c_k(i)\}_{k=1}^{3}$

Agent 1: Structure Guardian
Compute degree-change rate $\Delta_{\text{deg}}(i)$ and neighbor Jaccard similarity $J(i)$;
Flag banks by thresholds τ_{deg} and τ_J, and obtain confidence $c_1(i)$;
Agent 2: Feature Sentinel
Compare current intra-group correlations $\{\rho_g^{(t)}\}_{g=1}^{G_{\text{grp}}}$ with historical baselines (μ_g, σ_g), and flag large deviations with confidence $c_2(i)$;
Agent 3: Temporal Detective
Detect rating jumps $\Delta_r(i)$ unsupported by financial improvement $\phi(i)$ under thresholds τ_r and τ_ϕ, and obtain confidence $c_3(i)$;
Agent 4: Coordinator
Fuse agent confidences $\{c_k(i)\}_{k=1}^{3}$ into a fused suspiciousness score $s(i)$ via reliability-weighted voting;
Retain the top $m_{1_\text{top_percent}}$ positively scored nodes as S_1;
Optionally update agent reliability weights $\{w_k\}_{k=1}^{3}$;
return S_1, $\{s(i)\}_{i=1}^{N}$, and $\{c_k(i)\}_{k=1}^{3}$.

C. LLM-Enhanced Semantic Verification

Algorithm 3 LLM-Enhanced Semantic Verification

Input: candidate set S_1; financial feature matrix $X \in \mathbb{R}^{N \times d}$; adjacency matrix $A \in \mathbb{R}^{N \times N}$; ratings $\{y_i\}_{i=1}^{N}$ (optional); embedding node set $\mathcal{S}_{\text{emb}} \subseteq V$ (default: candidates in S_1)

Output: refined anomaly set S_2; semantic inconsistency score vector $\Omega \in \mathbb{R}^{N}$

Compute structure embeddings $Z^{\text{str}} \in \mathbb{R}^{N \times d_h}$ with a GCN on (X, A);
Serialize banks in $\mathcal{S}_{\text{emb}}$ with Prompt 1 and encode them with qwen3-embedding-8b to obtain semantic embeddings $Z^{\text{sem}} \in \mathbb{R}^{N \times d_s}$;
For nodes outside $\mathcal{S}_{\text{emb}}$, reuse Z^{str} as semantic placeholders;
Compute node-/type-/relation-level inconsistency diagnostics on $(Z^{\text{str}}, Z^{\text{sem}})$;
Fuse the diagnostics into Ω and retain high-Ω nodes within S_1, with a retention fallback if too few nodes remain;
return S_2 and Ω (zeroed outside S_2).

D. Attention-Guided Risk Propagation

Algorithm 4 Attention-Guided Risk Propagation

Input: semantic inconsistency score vector $\Omega \in \mathbb{R}^N$ and refined set S_2; adjacency matrix $A \in \mathbb{R}^{N\times N}$; rating histories $\{y_i^{(\tau)}\}$; optional causal diagnostics $(\Gamma_{\mathrm{Gr}}, \tilde{\Delta}_{\mathrm{cf}})$

Output: fused risk score vector $\psi \in \mathbb{R}^N$

Initialize risk field $r^{(0)} \in \mathbb{R}^N$ from Ω and S_2;
for $\ell = 1$ **to** L **do**
  $r^{(\ell)} \leftarrow$ DGARP step on $(r^{(\ell-1)}, A)$ with residual coefficient γ;
end for
(Optional) Compute STGCN temporal signals $s^{\mathrm{stgcn}} \in \mathbb{R}^N$ and Granger/SCM summaries $(\Gamma_{\mathrm{Gr}}, \Delta_{\mathrm{cf}})$;
Measure recent historical rating deterioration $\Delta r^{\mathrm{hist}} \in \mathbb{R}^N$;
Fuse components into $\psi \in \mathbb{R}^N$ and normalize to a common scale;
return ψ.

E. Statistical Attack Simulation

Algorithm 5 Statistical Attack Simulation

Input: period t; attack rate $\rho \in [0, 0.25]$

Output: attacked features $\tilde{X}^{(t)}$ and edges $\tilde{E}^{(t)}$

Feature Tampering
Estimate historical rating-conditional attribute means;
Sample a ρ-fraction of lower-rated banks;
Shift rating-3/4 attributes toward rating-1/2 means;
Edge Injection
Sample multi-stratum banks at rate ρ;
Pair higher-rated banks with lower-rated banks;
Append the induced cross-rating edges and store the tables;
return $\tilde{X}^{(t)}$, $\tilde{E}^{(t)}$.

II. Semantic Embedding Prompt and State Mapping

LLM-enhanced semantic verification uses a frozen embedding encoder. Prompt 1 wraps each structured financial profile so that credit-risk inconsistency is emphasized in embedding space. The same discrete state-label mapping is also reported in the main paper.

Prompt 1: Semantic Embedding Instruction

Role. Credit-risk inconsistency detector for interbank profiles.
Task. Encode a group-level financial summary so inconsistent profiles are separable in embedding space.
Input.
1) *Prefix:* [Credit rating anomaly: detect an inconsistent interbank profile.]
2) *Body:* [Bank <id>: Asset Scale, Capital Adequacy, Loan-Deposit Ratio, Profitability, Liquidity, and Risk Indicators, each as <state> (<value>).]
3) *Suffix:* [Anomalous vs. normal for rating.]

Constraint. Each <value> is the group mean and each <state> is its discrete label. The text is embedding input only, not a generative decision.

Discrete state labels are assigned by interval rules on the group-level mean. A value is labeled by the unique interval satisfying $\mathrm{lower} \le \mathrm{value} < \mathrm{upper}$; values outside all intervals are marked unknown. Table I lists each feature group, its state labels, and the corresponding intervals.

TABLE I
DISCRETE STATE-LABEL MAPPING FOR GROUP-LEVEL FINANCIAL MEANS.

Feature group	State	Interval
Asset / liability scale	high	$[10^{10}, +\infty)$
	medium	$[10^{9}, 10^{10})$
	low	$[0, 10^{9})$
Capital adequacy	sufficient	$[0.12, +\infty)$
	adequate	$[0.08, 0.12)$
	insufficient	$[0, 0.08)$
Loan-deposit	high	$[0.8, +\infty)$
	moderate	$[0.6, 0.8)$
	low	$[0, 0.6)$
Profitability	excellent	$[0.15, +\infty)$
	good	$[0.08, 0.15)$
	poor	$[0, 0.08)$
Liquidity	high	$[0.3, +\infty)$
	moderate	$[0.2, 0.3)$
	low	$[0, 0.2)$
Risk indicators	low	$[0, 0.02)$
	moderate	$[0.02, 0.05)$
	high	$[0.05, +\infty)$